1) $\left[\left(-\frac{1}{2}\right)^{-1}\right]^{-4} = -\left(-\frac{1}{2}\right)^{-4} =$

$= -\left(-\frac{1}{2}\right)^{-1-4} = -\left(-\frac{1}{2}\right)^{-4} = -2^4$

2) $-4^5 + 3 \cdot 4^5 - 7 \cdot 3^4 - 1 \cdot 3^4 - 4^5 =$

$4^5(-1+3-1) + 3^4(-7-3)$

$4^5 \cdot 1 + 3^4 \cdot (-10)$

$= 4^5 - 10 \cdot 3^4$

3- $\frac{5^{80} + 5^{85} + 5^{90}}{5^{85} + 5^{80} + 5^{75}} = x$ $\quad \frac{x}{125} = ?$

$\frac{5^{80} + 5 \cdot 5^{80} + 25 \cdot 5^{80}}{5^{75} + 5 \cdot 5^{80} + 25 \cdot 5^{80}} = \frac{5^{80}}{5^{75}} = 5^{80-75} = 5^5$

$x = 5^5$

$\frac{x}{125} = \frac{5^5}{5^3} = 5^{5-3} = 5^2 = 25$

4- $\frac{14^n + 6^n + 2^n}{7^n + 3^n + 1} = \frac{7^n \cdot 2^n + 3^n + 2^n + 2^n}{7^n + 3^n + 1}$

$14^n = (7 \cdot 2)^n = 7^n \cdot 2^n$

$6^n = (3 \cdot 2)^n = 3^n \cdot 2^n$

$2^n = 1 \cdot 2^n$

$\frac{7^n + 3^n + 1}{2^n(7^n + 3^n + 1)} = \frac{7^n + 3^n + x}{} = 2^n$

$3-\frac{5}{3}$

$(0,027)^{\frac{5}{3}} \cdot 10^5$

$\frac{27}{1000} = \left(\frac{3}{10}\right)^3 \left(\left(\frac{3}{10}\right)^3\right)^{\frac{5}{3}} \cdot 10^5$

$\left(\frac{3}{10}\right)^5 \cdot 10^5 = \frac{3^5}{10^5} \cdot 10^5 = 3^5$

$$\frac{-9^{-2} \cdot 25^{-1}}{125^{-1} \cdot 27^{-1}} = ?$$

$$\frac{-(3^2)^{-2} \cdot (5^2)^{-1}}{(5^3)^{-1} \cdot (3^3)^{-1}} = \frac{-3^{-4} \cdot 5^{-2}}{5^{-3} \cdot 3^{-3}} = \frac{5^{3-2}}{3^{4-3}} = \frac{-5}{3}$$

$$\frac{x^{a+1}}{x^{a-2}} \left(\frac{x^{-a+2} \cdot x^{2a+3}}{x} \right)^{-1}$$

$$x^{(a+1)-(a-2)} \cdot \left(x^{-a+2-2a+3-1}\right)^{-1} =$$

$$x^3 \cdot \frac{\left(x^{a+4}\right)^{-1}}{x}$$

$$= x^3 \cdot x^{-a-4} = x^{3-a-4} = x^{-a-1}$$

$$\frac{1}{x^{a+1}}$$

✱ $\left(\frac{a^x}{a^y}\right)^{x-y} - \left(\frac{a^y}{a^x}\right)^{x-y}$

$(a^{x-y})^{x-y} , (a^{y-x})^{x-y} = (a^{x-y} \cdot a^{y-x})^{x-y} -$

$x-y + y-x$

a

$(a^0)^{x-y} = 1^{x-y} = 1$

$\overline{1}$

P $3^x = 2$ $\quad 27^{x+1} - 81^x$

$(3^3)^{x+1} - (3^4)^x = 3^{3x+3} - 3^{4x}$

$= 3^{3x} \cdot 3^3 = (3^x)^3 \cdot 27$

$(3^x)^3 \cdot 3^3 - (3^x)^4$

$2^3 \cdot 3^3 - 2^4$

$= 100$

$$\frac{\sqrt{75}-\sqrt{27}+\sqrt{12}}{\sqrt{108}+\sqrt{48}} = \frac{\sqrt{5^2\cdot 3}-\sqrt{3^2\cdot 3}+\sqrt{2^2\cdot 3}}{\sqrt{6^2\cdot 3}+\sqrt{4^2\cdot 3}}$$

$$= \frac{5\sqrt{3}-3\sqrt{3}+2\sqrt{3}}{6\sqrt{3}+4\sqrt{3}}$$

$$= \frac{4\sqrt{3}}{10\sqrt{3}} = \frac{4}{10} = \frac{2}{5}$$

$\sqrt{2}\cdot(\sqrt{18}+\sqrt{50}-\sqrt{32}+\sqrt{8}=$

$\sqrt{18}=\sqrt{9\cdot2}=3\sqrt{2}$

$\sqrt{50}=\sqrt{25\cdot2}=5\sqrt{2}$

$\sqrt{32}=\sqrt{16\cdot2}=4\sqrt{2}$

$\sqrt{8}=\sqrt{4\cdot2}=2\sqrt{2}$

$3\sqrt{2}+5\sqrt{2}-4\sqrt{2}+2\sqrt{2}$

$(3+5-4+2)\sqrt{2}=6\sqrt{2}$

$2\cdot6\sqrt{2}=6\cdot(\sqrt{2}\cdot\sqrt{2})=6\cdot2\cdot2$

$5\sqrt{\frac{16}{\frac{1}{2}}}=5\sqrt{\frac{16}{1}\cdot2}=5\sqrt{32}=5\sqrt{2^5}=$

$=2$

$\sqrt{25-25x^2}-\sqrt{64-64x^2}=$

$\sqrt{25(1-x^2)}-\sqrt{64-(1-x^2)}$

$5\sqrt{1-x^2}-8\sqrt{1-x^2}$

$5-3\sqrt{1-x^2}=-3\sqrt{1-x^2}$

$\sqrt[3]{2\sqrt[3]{4}}=$

$\sqrt{(-4)^2} - \sqrt[3]{(-2)^3} + \sqrt[4]{(-2)^4} + \sqrt[5]{3^5} =$

$\sqrt{(-4)^2} = |-4| = 4$

$\sqrt[3]{(-2)^3} = -2$

$\sqrt[4]{(-2)^4} = |-2| = 2$ çift'

$\sqrt[5]{3^5} = 3$ (Tek)

$4 - (-2) = 4 + 2 = 6$

$6 + 2 = 8$

$8 + 3 = 11$

$\frac{2}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$

$\sqrt{6}$ ile sadeleşir